

ChE 312 – Transport Processes II

III. Separations: **Absorption / Stripping / (Humidification).**

Transfer a component from gas to liquid (absorption) or from liquid to gas (stripping).

Previously: Absorption/stripping with equilibrium stages.

Continued today: Continuous countercontacting within packed columns.

Absorber/stripper design by integrating the continuous mass transfer.

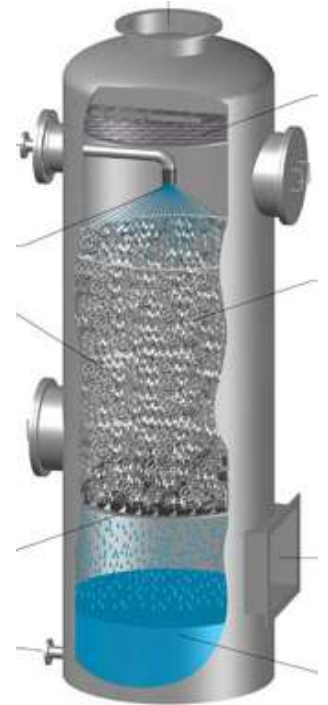
*****Phase equilibrium at the interface!!!*****

Gives stripper / absorber packed-column equations to be integrated.

“Transfer-units” approach; analytical methods and $z=NTU*HTU$.

Continuous-contacting design.

Examine packings, pressure drop, sizing the diameter to avoid flooding.



**Example 22.7-1 is a comprehensive example,
which you'll re-work in PS16.**

EXAMPLE 22.7-1. Design of an Absorption Tower with a Concentrated Gas Mixture

A tower packed with 25.4-mm ceramic rings is to be designed to absorb SO_2 from air by using pure water at 293 K and 1.013×10^5 Pa abs pressure. The entering gas contains 20 mol % SO_2 and that leaving contains 2 mol %. The inert air flow is 6.53×10^{-4} kg mol air/s and the inert water flow is 4.20×10^{-2} kg mol water/s. The tower cross-sectional area is 0.0929 m^2 . For dilute SO_2 , the film mass-transfer coefficients at 293 K are, for 25.4-mm (1-in.) rings (W1),

$$k'_y a = 0.0594 G_y^{0.7} G_x^{0.25} \quad k'_x a = 0.152 G_x^{0.82}$$

where $k'_y a$ is $\text{kg mol/s} \cdot \text{m}^3 \cdot \text{mol frac}$, $k'_x a$ is $\text{kg mol/s} \cdot \text{m}^3 \cdot \text{mol frac}$, and G_x and G_y are kg total liquid or gas, respectively, per sec per m^2 tower cross section. Calculate the tower height.

Step 1: Sketch of what you know and what you don't.

In air: $y_2=2\%$ SO₂
 V' unchanging

$x_2=0\%$ SO₂ in H₂O;
 $L'=4.20 \cdot 10^{-2}$ kmol water /s
 ("inert")

$T=293\text{K}$, $P=1.013 \cdot 10^5$ Pa;
 25.4mm ceramic rings;
 Tower CSA= 0.0929 m^2 ;
 $k'_y a = 0.0594 G_y^{0.7} G_x^{0.25}$
 $k'_x a = 0.152 G_x^{0.82}$
 with
 $k'a$ in kmol/(s·m³·mol frac);
 G in kg·s⁻¹·m⁻²

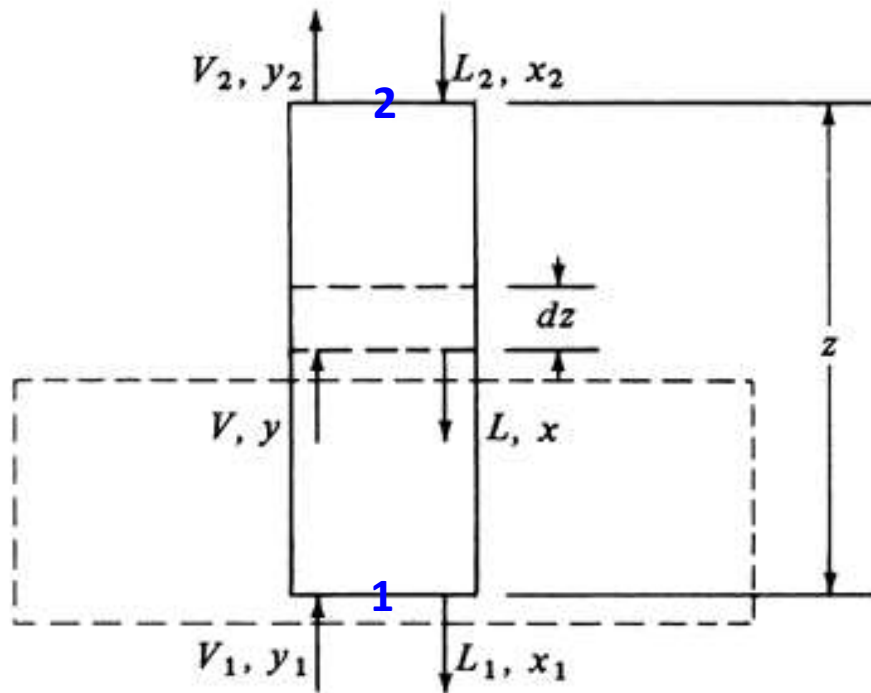
Packing height z ?

In air: $y_1=20\%$ SO₂
 $V'=6.53 \cdot 10^{-4}$ kmol/s air
 ("inert")

$x_1=?\%$ SO₂ in water
 L' unchanging



Step 2: Next, apply overall balance to get x_1 .



Total flows: $L = \frac{L'}{1-x}$ and $V = \frac{V'}{1-y}$

Overall SO_2 : $L_2 x_2 + V_1 y_1 = L_1 x_1 + V_2 y_2$

$$\frac{L'}{1-x_2} x_2 + \frac{V'}{1-y_1} y_1 = \frac{L'}{1-x_1} x_1 + \frac{V'}{1-y_2} y_2$$

[Also, operating line gives x and y SO_2 at any height z :]

$$\begin{aligned} & L' \left(\frac{x}{1-x} \right) + V' \left(\frac{y_1}{1-y_1} \right) \\ &= L' \left(\frac{x_1}{1-x_1} \right) + V' \left(\frac{y}{1-y} \right) \end{aligned}$$

Next, apply overall balance to get x_1 .

In air: $y_2=2\%$ SO₂
 V' unchanging

$x_2=0\%$ SO₂ in H₂O;
 $L'=4.20 \cdot 10^{-2}$ kmol water /s
 ("inert")

$T=293\text{K}$, $P=1.013 \cdot 10^5$ Pa;
 25.4mm ceramic rings;
 Tower CSA=0.0929 m²;
 $k'_y a = 0.0594 G_y^{0.7} G_x^{0.25}$
 $k'_x a = 0.152 G_x^{0.82}$
 with
 $k'a$ in kmol/(s·m³·mol frac);
 G in kg·s⁻¹·m⁻²

Packing height z ?

In air: $y_1=20\%$ SO₂
 $V'=6.53 \cdot 10^{-4}$ kmol/s air
 ("inert")

$x_1=?\%$ SO₂ in water
 L' unchanging

Total flows: $L = \frac{L'}{1-x}$ and $V = \frac{V'}{1-y}$

Overall SO₂: $L_2 x_2 + V_1 y_1 = L_1 x_1 + V_2 y_2$

$$\frac{L'}{1-x_2} x_2 + \frac{V'}{1-y_1} y_1 = \frac{L'}{1-x_1} x_1 + \frac{V'}{1-y_2} y_2$$

Solving

$$\begin{aligned} & 4.20 \cdot 10^{-2} \left(\frac{0}{1-0} \right) + 6.53 \cdot 10^{-4} \left(\frac{0.20}{1-0.20} \right) \\ &= 4.20 \cdot 10^{-2} \left(\frac{x_1}{1-x_1} \right) + 6.53 \cdot 10^{-4} \left(\frac{0.02}{1-0.02} \right) \end{aligned}$$

gives $x_1 = 0.00355$

Step 3: Use the operating-line equation (mass balance).

In Air:
 $y_2=2\%$ SO₂

$x_2=0\%$ SO₂ in H₂O;
 $L'=4.20 \cdot 10^{-2}$ kmol/s
 water ("inert")



In air:
 $y_1=20\%$ SO₂
 $V'=6.53 \cdot 10^{-4}$
 kmol/s air
 ("inert")

$x_1=0.355\%$ SO₂
 in water

Relate mole fractions x and y SO₂ at any height z :

$$\begin{aligned}
 & L' \left(\frac{x}{1-x} \right) + V' \left(\frac{y_1}{1-y_1} \right) \\
 &= L' \left(\frac{x_1}{1-x_1} \right) + V' \left(\frac{y}{1-y} \right) \\
 & 4.20 \cdot 10^{-2} \left(\frac{x}{1-x} \right) + 6.53 \cdot 10^{-4} \left(\frac{0.20}{1-0.20} \right) \\
 &= 4.20 \cdot 10^{-2} \left(\frac{0.00355}{1-0.00355} \right) + 6.53 \cdot 10^{-4} \left(\frac{y}{1-y} \right)
 \end{aligned}$$

Step 4: Compose the differential MT equation for any z.

$$d(Vy) = \frac{V dy_{AG}}{1 - y_{AG}} = N_A dA = N_A \times a \times S \times dz = \frac{k_y a}{(1 - y_A)_{iM}} (y_{AG} - y_{Ai}) \times S \times dz$$

$$d(Lx) = \frac{L dx_{AL}}{1 - x_{AL}} = N_A dA = N_A \times a \times S \times dz = \frac{k_x a}{(1 - x_A)_{iM}} (x_{Ai} - x_{AL}) \times S \times dz$$

$$dz = \frac{V \cdot dy_A}{1 - y_A} \cdot \frac{1}{\frac{k'_y a}{(1 - y_A)_{iM}} (y_A - y_{Ai}) \cdot S} = \frac{V}{\frac{k'_y a \cdot S}{(1 - y_A)_{iM}} (1 - y_A) (y_A - y_{Ai})} dy_A$$

- Strategy: For each in a series of x values, get y and tabulate.
- At each position, get and tabulate $V = V'/(1-y)$
- Likewise L, G_y , G_x to get $k'_y a$ and $k'_x a$.

Step 5: Get x_i, y_i interface values.

- Using $k'_y a$ and $k'_x a$ with x, y and trial x_i, y_i , get new estimates of x_i, y_i using:

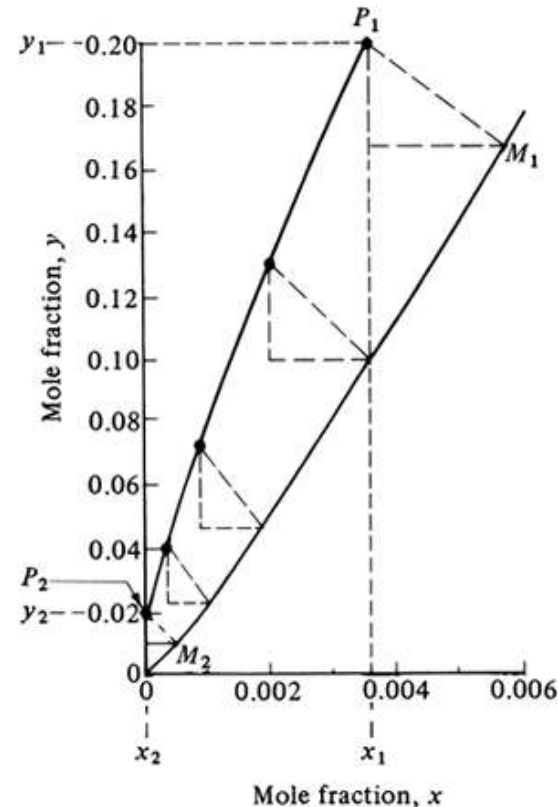
$$\text{slope}(PM) = \frac{y_i - y}{x_i - x} = - \frac{k'_x a / (1 - x)_{iM}}{k'_y a / (1 - y)_{iM}}$$

where $y_i = y_{\text{equil}}(x_i)$

and find x_i, y_i for each x, y by trial and error.

- Or solve for x_i with Solver or *fzero*:

$$\frac{y_{\text{equil}}(x_i) - y}{x_i - x} + \frac{k'_x a / (1 - x)_{iM}}{k'_y a / (1 - y)_{iM}} = 0$$



Step 6: Integrate numerically using the collected numbers:

$$d(Vy) = \frac{V dy_{AG}}{1 - y_{AG}} = N_A dA = N_A a S dz = \frac{k'_y a S}{(1 - y_A)_{iM}} (y_{AG} - y_{Ai}) S dz$$

$$d(Lx) = \frac{L dx_{AL}}{1 - x_{AL}} = N_A dA = N_A a S dz = \frac{k'_x a S}{(1 - x_A)_{iM}} (x_{Ai} - x_{AL}) S dz$$

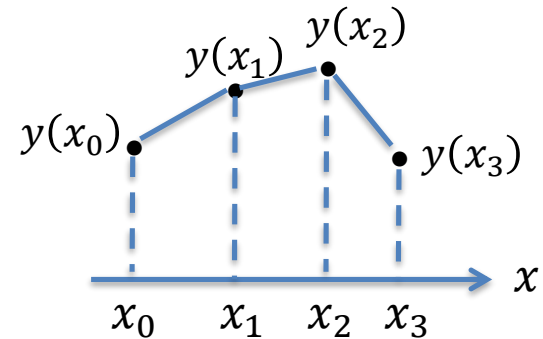
$$z = \int_{y_2}^{y_1} \left[\frac{V}{\frac{k'_y a S}{(1 - y_i)_{iM}} (1 - y)(y - y_i)} \right] dy$$

													$\frac{V}{\frac{k'_y a S}{(1 - y)_{iM}} (1 - y)(y - y_i)}$
y	x	V	L	G _y	G _x	k' _y a	k' _x a	x _i	y _i	1-y	(1-y) _{iM}	y - y _i	
0.02	0	6.66 × 10 ⁻⁴	0.04200	0.2130	8.138	0.03398	0.848	0.00046	0.0090	0.980	0.985	0.0110	19.25
0.04	0.000332	6.80 × 10 ⁻⁴	0.04201	0.2226	8.147	0.03504	0.849	0.00103	0.0235	0.960	0.968	0.0165	12.77
0.07	0.000855	7.02 × 10 ⁻⁴	0.04203	0.2378	8.162	0.03673	0.850	0.00185	0.0476	0.930	0.941	0.0224	9.29
0.13	0.00201	7.51 × 10 ⁻⁴	0.04208	0.2712	8.196	0.04032	0.853	0.00355	0.1015	0.870	0.885	0.0285	7.16
0.20	0.00355	8.16 × 10 ⁻⁴	0.04215	0.3164	8.241	0.04496	0.857	0.00565	0.1685	0.800	0.816	0.0315	6.33

Numerical integration: $\int y(x) \cdot dx \cong \sum \left\{ \frac{y(x_i) + y(x_{i+1})}{2} \cdot (x_{i+1} - x_i) \right\}$

- Easiest numerical approach is to use trapezoidal increments on x values:

- $(\text{Average of } y(x_0) \text{ and } y(x_1)) \cdot (x_1 - x_0)$
- Plus $(\text{Average of } y(x_1) \text{ and } y(x_2)) \cdot (x_2 - x_1)$
- Plus $(\text{Average of } y(x_2) \text{ and } y(x_3)) \cdot (x_3 - x_2)$
- And so on to the end.



- The x_n values don't have to be equally spaced; the smaller Δx is, the better.
- In MATLAB, you can use ode solvers.
- Or you can program numerical methods like Runge-Kutta.

Observe the general form of the balance equations, and use it.

$$d(Vy) = \frac{V dy_{AG}}{1 - y_{AG}} = N_A dA = N_A \cdot a \cdot S \cdot dz = \frac{k'_y a}{(1 - y_A)_{iM}} (y_{AG} - y_{Ai}) \cdot S \cdot dz$$

$$\begin{aligned} \int dz &= \int \frac{V \cdot dy_A}{1 - y_A} \cdot \frac{1}{\frac{k'_y a}{(1 - y_A)_{iM}} (y_A - y_{Ai}) \cdot S} \\ &= \int \frac{V}{\frac{k'_y a \cdot S}{(1 - y_A)_{iM}} (1 - y_A) (y_A - y_{Ai})} dy_A = \frac{V}{k'_y a \cdot S} \cdot \int \frac{(1 - y_A)_{iM} \cdot dy_A}{(1 - y_A) (y_A - y_{Ai})} \end{aligned}$$

$$\begin{aligned} \text{Units: } & \frac{\text{mol/s}}{\left(\frac{\text{mol/s}}{\text{m}^2 \text{mol frac}} \right) \left(\frac{\text{m}^2}{\text{m}^3} \right) \cdot \text{m}^2} = m \\ & \equiv \text{"Height of a transfer unit"} \end{aligned}$$

“Number of transfer units”

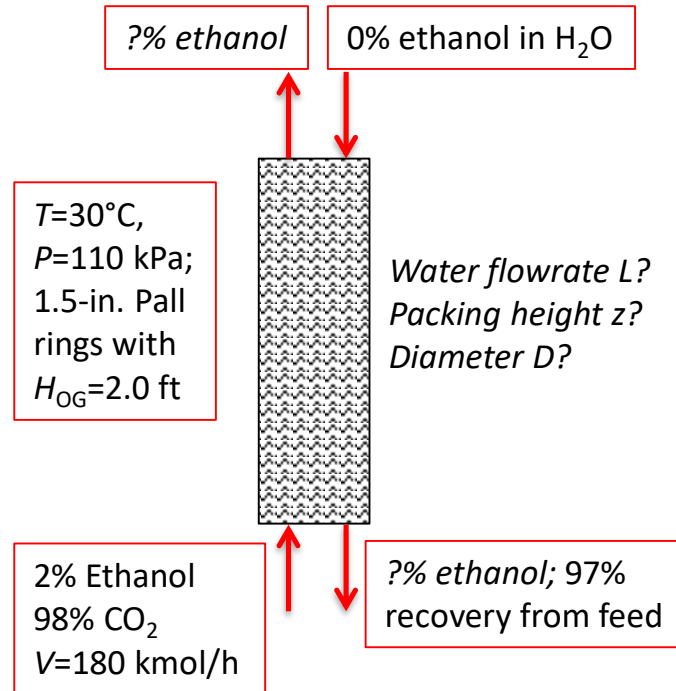
“What ifs”: What variations can happen?

- Calculate pinch point (flow that causes op line and eq line to intersect between x_1, y_1 and x_2, y_2) and estimate optimal flow.
- Base the integrand on $k'_x a$ if dominant resistance is in the liquid.
 - Or overall $K'_x a$ or $K'_y a$.
- Different packings and packing density.
- Use flooding velocities to find diameter.

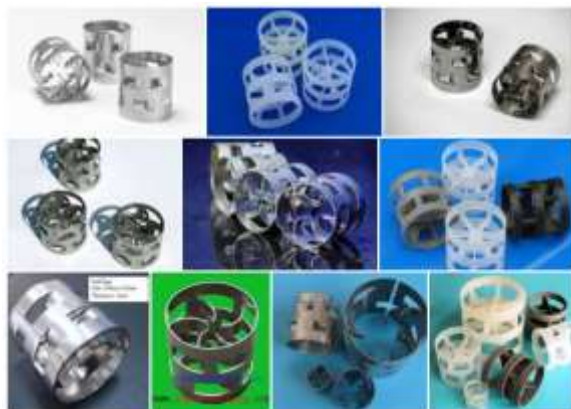
Another absorber example.

- In a fermentation process, ethanol is being produced from cane-sugar extract.
- Part of the product is ethanol in a CO₂-rich gas stream: approximately 180 kmol/hr of 2 mol % ethanol (“A”), 98% CO₂, at 30°C and 110 kPa.
 - [Because these conditions are nominal design values, we can treat them as exact numbers and recognize their limits.]
- Countercurrent absorption into water (also at 30°C and 110 kPa) is to be used for 97% ethanol recovery.
- The packed column uses 1.5” Pall rings with $H_{OG}=2.0$ ft.
- **Design the process.**

Step 1: Sketch what you know.



Get info about the packing.



[More images for pall ring](#)

[The Pall Ring Company - Home](#)

www.pallrings.co.uk/

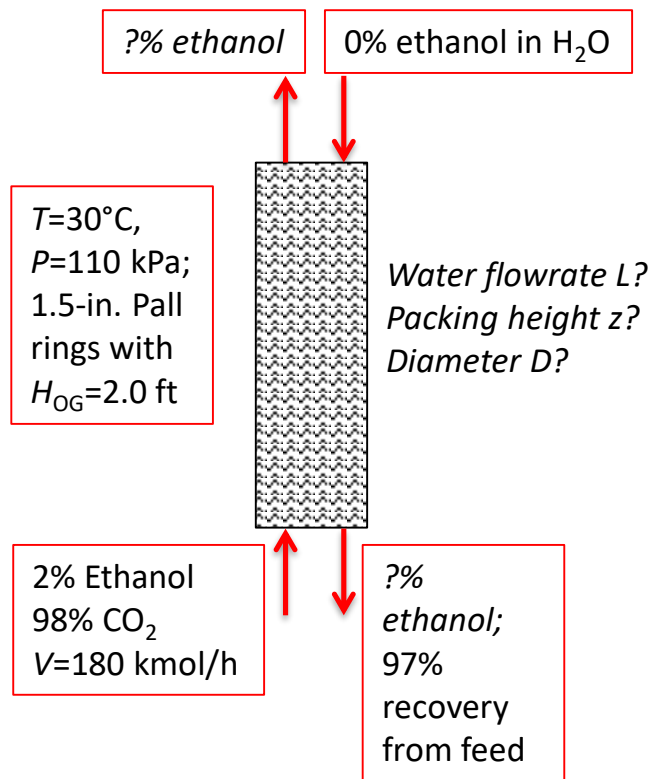
The Pall Ring Company provides products for mass transfer operations including plastic and metal tower packings.

Table 22.3-1. Packing Factors for Random and Structured Packing

Type	Material	Nominal size, in.	Void fraction, ϵ	Surface area, a , ft^2/ft^3 (m^2/m^3)	Packing factor, F_p , ft^{-1} (m $^{-1}$)	Relative mass-transfer coefficient, f_p
Random Packing						
Raschig Rings	Ceramic	1/2	0.64	111 (364)	580 (1900)	1.52
		1	0.74	58 (190)	179 (587)	1.20
		1 1/2	0.73	37 (121)	95 (312)	1.00
		2	0.74	28 (92)	65 (213)	0.85
Berl Saddles	Ceramic	1/2	0.62	142 (466)	240 (787)	1.58
		1	0.68	76 (249)	110 (361)	1.36
		2		32 (105)	45 (148)	
Pall Rings	Metal	1	0.94	63 (207)	56 (184)	1.61
		1 1/2	0.95	39 (128)	40 (131)	1.34
		2	0.96	31 (102)	27 (89)	1.14
Metal Intalox (IMTP)	Metal	1	0.97	70 (230)	41 (134)	1.78
Nor-Pac	Plastic	2	0.98	30 (98)	18 (59)	1.27
		1	0.92	55 (180)	25 (82)	
Hy-Pak	Metal	2	0.94	31 (102)	12 (39)	
		1	0.96	54 (177)	45 (148)	1.51
	Plastic	2	0.97	29 (95)	26 (85)	1.07
		1	0.92	55 (180)	25 (82)	
		2	0.94	31 (102)	12 (39)	
Structured Packing						
Mellapak 250Y	Metal		0.95	76 (249)	20 (66)	
		500Y		152 (499)	34 (112)	
Flexipac		2	0.93	68 (223)	22 (72)	
		4	0.98		6 (20)	
Gempak		2A	0.93	67 (220)	16 (52)	
		4A	0.91	138 (452)	32 (105)	
Norton Intalox		2T	0.97	65 (213)	17 (56)	1.98
		3T	0.97	54 (177)	13 (43)	1.94
Montz B300				91 (299)	33 (108)	
Sulzer	Wire Mesh	CY	0.85	213 (700)	70 (230)	
		BX	0.90	150 (492)	21 (69)	

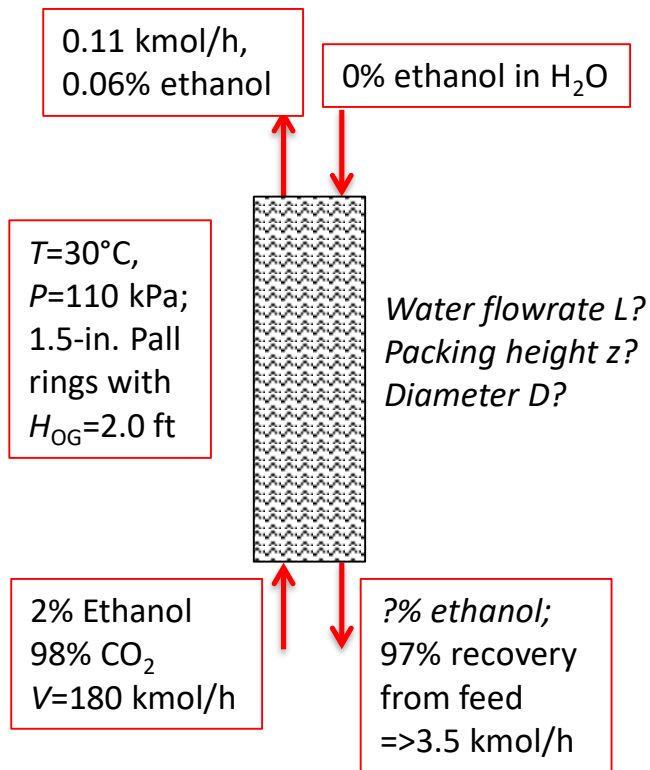
Data from Ref. (K1, L2, P2, S4). The relative mass-transfer coefficient, f_p , is discussed in [Section 22.8B](#).

Step 2: Apply overall balances.



- “97% recovery” means 97% of the fed ethanol ($0.02 \times 180 \text{ kmol/h}$) is absorbed; i.e., 3.5 kmol/h ethanol in the exiting H_2O .
- That determines the ethanol in the exiting gas: $0.03(0.02 \times 180 \text{ kmol/h}) = 0.11 \text{ kmol/h}$, or $y = 0.11 / (180 - 3.5) = 0.0006$
- L to be determined

Step 3: Determine $(L'/V')_{\text{minimum}}$



- $(L'/V')_{\text{minimum}}$ where operating line intersects equil line; pinch pt.
- Equil line is $y_A P = \gamma_A x_A P_{\text{vap},A}$
 - At 30°C, $P_{\text{vap},\text{EtOH}} = 10.5 \text{ kPa}$.
 - From infinite-dilution data, $\gamma_{\text{EtOH}} = 6$.
 - So equilibrium line is $y_A = 0.57 x_A$.
- Operating line

$$L' \frac{x_1}{1-x_1} + V' \frac{y_1}{1-y_1} = L' \frac{x_2}{1-x_2} + V' \frac{y_2}{1-y_2}$$

goes through $(x_1, y_1) = (0, 0.0006)$

- Solve for intersection of the equilibrium and operating-line equations, such as with Excel Goal Seek or Solver.

	A	B	C	D	E	F	G	H	I
1	$\left(\frac{y}{1-y}\right) = \frac{L'}{V'} \left(\frac{x}{1-x} - \frac{x_1}{1-x_1}\right) + \left(\frac{y_1}{1-y_1}\right)$ $\frac{L'}{V'} \left(\frac{x}{1-x}\right) = \left(\frac{y}{1-y}\right) - \left(\frac{0.0006}{0.9994}\right)$ $\frac{L'}{V'} x = \left(\frac{y}{1-y}\right)(1-x) - \left(\frac{0.0006}{0.9994}\right)(1-x)$ $x = \frac{\left(\frac{y}{1-y}\right) - \left(\frac{0.0006}{0.9994}\right)}{\frac{L'}{V'} - \left(\frac{0.0006}{0.9994}\right) + \left(\frac{y}{1-y}\right)} = \left(\frac{y}{0.57}\right)_{\text{equilibrium}}$ so $0 = \frac{\left(\frac{y}{1-y}\right) - \left(\frac{0.0006}{0.9994}\right)}{\frac{L'}{V'} - \left(\frac{0.0006}{0.9994}\right) + \left(\frac{y}{1-y}\right)} - \frac{y}{0.57}$ or $y = \frac{\frac{L'}{V'} \left(\frac{x}{1-x}\right) + \left(\frac{0.0006}{0.9994}\right)}{\left(1 + \frac{L'}{V'} \left(\frac{x}{1-x}\right) + \frac{0.0006}{0.9994}\right)} \xrightarrow{\text{equil}} 0.57x$								
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$$f(L/V, y) = (E5/(1-E5) - 0.0006/0.9994) / (D3 - (0.0006/0.9994) + E5/(1-E5)) - E5/0.57$$

$$0.0002806$$

$$y_2 = 0.02$$

$$L/V = 0.54023579$$

Goal Seek

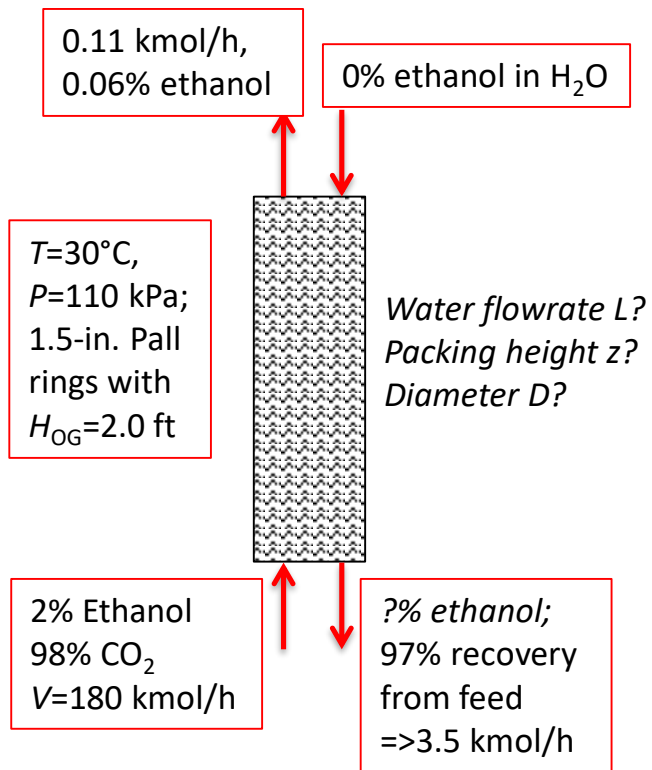
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Step 4: Use $1.5(L'/V')_{\text{minimum}}$



- $1.5(L'/V')_{\text{optimum}} = 1.5(L'/V')_{\text{minimum}} = 1.5 \times 0.540 = 0.81$
- $V' = 0.98 \times 180 = 176\text{ kmol/h}$, so $L' = 143\text{ kmol/h}$.
- Now you can determine x_2 : $3.5 / (3.5 + 143) = 0.024$ or 2.4% and $L = 147\text{ kmol/h}$.
- Continue as in previous example to integrate the mass-transfer equation; then multiply by H_{OG} for height z .

Flooding is an operational problem to avoid, but it is a useful reference for design.

- We discussed flooding before in distillation.
- If L/V is too low, liquid will back up and not flow down, stopping operation.
 - Gas flow is too high if (pressure drop)x(cross sectional area) exceeds the weight of the liquid: $(\Delta p_{packed\ bed})(S) > mg \cdot \Delta h$
- However, knowing the flooding velocity allows the designer to find a limiting cross-sectional area (or limiting diameter) at which flooding would occur.
 - Then make the diameter safely bigger.
- The key is calculating pressure drop.

Use flooding velocities to find diameter.

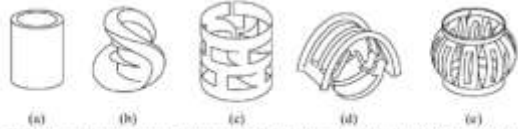


Figure 22.3-4. Typical random or dumped tower packings: (a) Raschig ring; (b) Berl saddle; (c) Pall ring; (d) Intalox metal; (e) Jaeger Metal Tri Pack.

capacity parameter $v_G [\rho_G / (\rho_L - \rho_G)]^{0.5} F_p^{0.5} \nu^{0.05}$

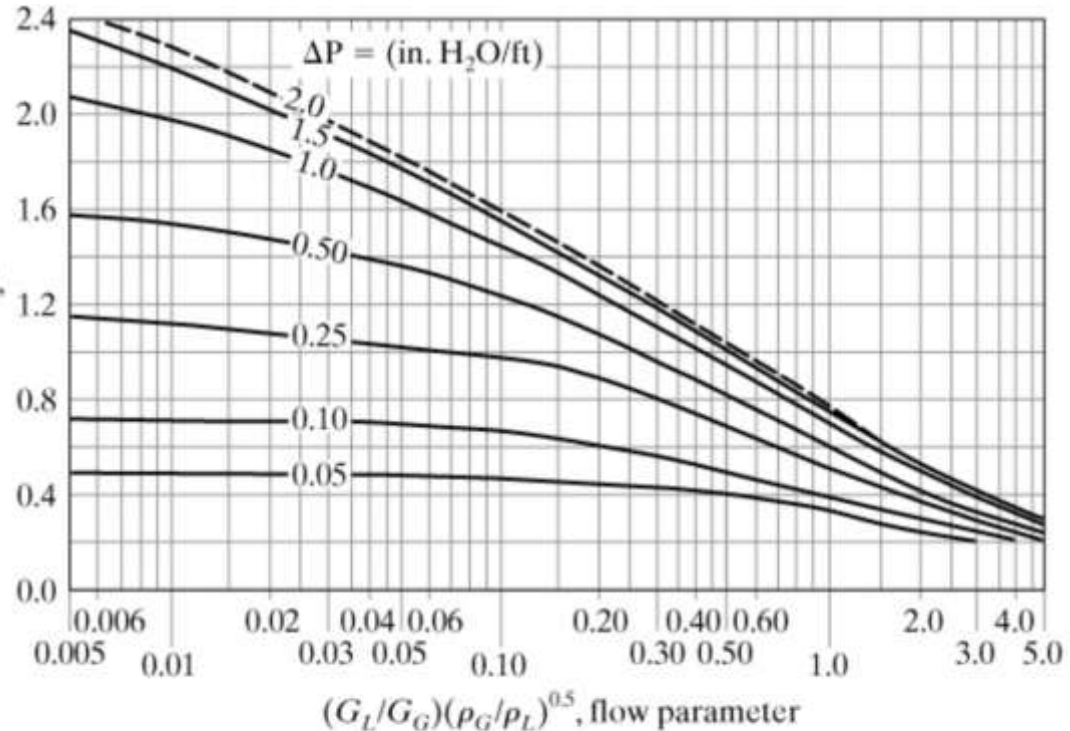


Figure 22.3-1. Pressure-drop correlation for random packings by Strigle. Data from R. F. Strigle, Jr., *Random Packings and Packed Towers*, © 1987, Houston: Gulf Publishing Company.

Similar correlation for structured packings.

capacity parameter $v_G[\rho_G/(\rho_L - \rho_G)]^{0.5} F_p^{0.5} \nu^{0.05}$

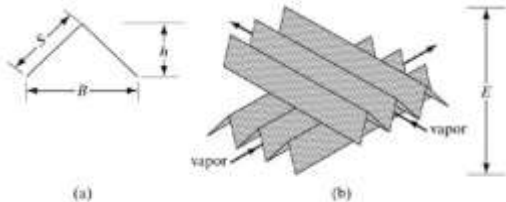


Figure 10 B-2. Typical corrugated structured packing: (a) triangular cross section of flow channel; (b) flow-channel arrangement, with vapor flowing upward. Indicated by arrows, and liquid downward. [From J. R. Fair and J. L. Glass, *Chem. Eng. Progr., 86*, 54M, 1, 11 (1986). With permission.]

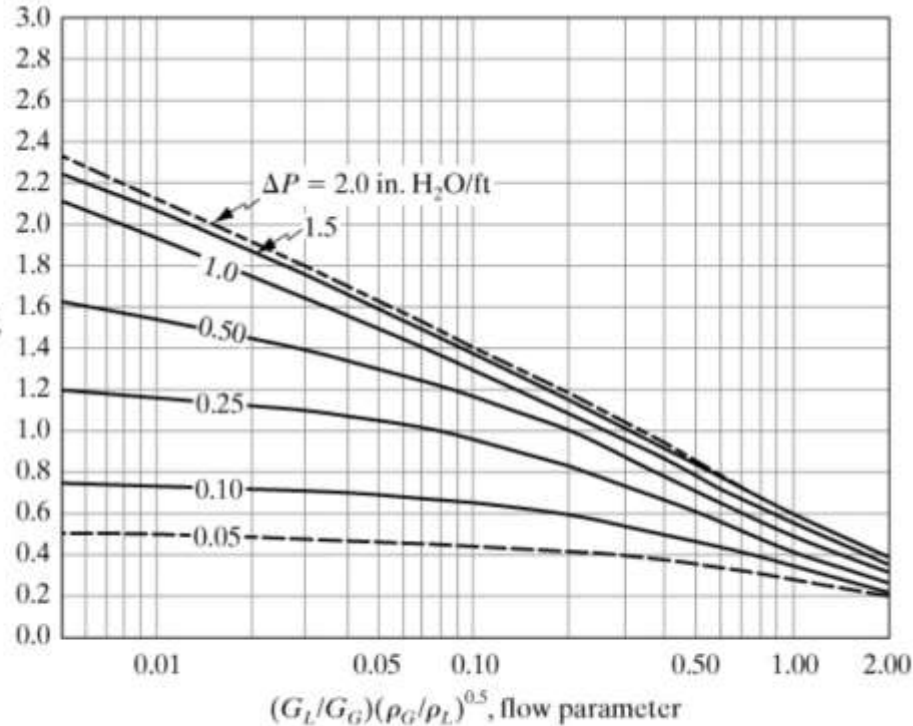


Figure 22.3-2. Pressure-drop correlation for structured packings by Kister and Gill (K2). From H. Z. Kister, *Distillation Design*, New York, McGraw-Hill Book Company, 1992. With permission.

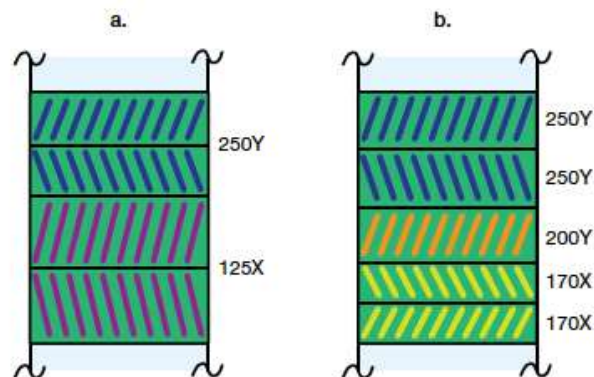
Table 22.3-1 compares packing parameters.

Note size,
void fraction,
interfacial area a ,
 F_p (important for
flooding
correlation),
 f_p (important for
mass transfer
coefficient
correlations).

Type	Material	Nominal size, in.	Void fraction, ϵ	Surface area, a , ft^2/ft^3 (m^2/m^3)	Packing factor, F_p , ft^{-1} (m^{-1})	Relative mass transfer coefficient, f_p
<i>Random Packing</i>						
Raschig rings	Ceramic	1/2	0.64	111 (364)	580 (1900)	1.52
		1	0.74	58 (190)	179 (587)	1.20
		1 1/2	0.73	37 (121)	95 (312)	1.00
		2	0.74	28 (92)	65 (213)	0.85
Berl saddles	Ceramic	1/2	0.62	142 (466)	240 (787)	1.58
		1	0.68	76 (249)	110 (361)	1.36
		2		32 (105)	45 (148)	
Pall rings	Metal	1	0.94	63 (207)	56 (184)	1.61
		1 1/2	0.95	39 (128)	40 (131)	1.34
		2	0.96	31 (102)	27 (89)	1.14
Metal Intalox (IMTP)	Metal	1	0.97	70 (230)	41 (134)	1.78
		2	0.98	30 (98)	18 (59)	1.27
Nor-Pac	Plastic	1	0.92	55 (180)	25 (82)	
		2	0.94	31 (102)	12 (39)	
Hy-Pak	Metal	1	0.96	54 (177)	45 (148)	1.51
		2	0.97	29 (95)	26 (85)	1.07
	Plastic	1	0.92	55 (180)	25 (82)	
		2	0.94	31 (102)	12 (39)	
<i>Structured Packing</i>						
Mellapak	250Y	Metal	0.95	76 (249)	20 (66)	
	500Y		152 (499)	34 (112)		
Flexipac	2		0.93	68 (223)	22 (72)	
	4		0.98	6 (20)		
Gempak	2A		0.93	67 (220)	16 (52)	
	4A		0.91	138 (452)	32 (105)	
Norton Intalox	2T		0.97	65 (213)	17 (56)	1.98
	3T		0.97	54 (177)	13 (43)	1.94
Montz	B300		91 (299)	33 (108)		
Sulzer	CY	Wire	0.85	213 (700)	70 (230)	
		Mesh				
	BX		0.90	150 (492)	21 (69)	

Data from Ref. (K1, L2, P2, S4). The relative mass-transfer coefficient, f_p , is discussed in Section 22.8B.

Aids to fixes and improvements:



▲ **Figure 5.** a. A vendor proposed replacing a portion of packing in a packed bed with more-open packing to accommodate high vapor and liquid loadings at the bottom of the bed. However, this would initiate flooding in the inlet to the upper, higher-efficiency section. b. A three-size packing design optimized both heat transfer and capacity in the packed bed. (The generic use of X and Y in the packing size descriptions designates the two typical angles of the crimps and resulting vapor channels in the packing. X designates structured packing with a crimp and vapor channel that is 60-deg. to the horizontal; Y designates a 45-deg. crimp and vapor channel. Thus, 125X packing has a surface area of $125 \text{ m}^2/\text{m}^3$ with a crimp angle of 60 deg.)



▲ **Figure 6.** A mini packing was used in a 6-in. column. However, oil buildup on the small packing elements reduced the open area of packing significantly, causing flooding. Photo courtesy of Cannon Instruments Co.

Apply coefficients to design separators.

$$z = H_G \int_{y_2}^{y_1} \frac{(1-y)_{iM} dy}{(1-y)(y-y_i)} = H_G N_G \quad H_G = \frac{V}{k'_y a S} = \frac{V}{k_y a (1-y)_{iM} S} \quad z = H_G N_G = H_G \int_{y_2}^{y_1} \frac{dy}{y-y_i}$$

or

$$z = H_L \int_{x_2}^{x_1} \frac{(1-x)_{iM} dx}{(1-x)(x_i-x)} = H_L N_L \quad H_L = \frac{L}{k'_x a S} = \frac{L}{k_x a (1-x)_{iM} S} \quad z = H_L N_L = H_L \int_{x_2}^{x_1} \frac{dx}{x_i-x}$$

or

$$z = H_{OG} \int_{y_2}^{y_1} \frac{(1-y)_{*M} dy}{(1-y)(y-y^*)} = H_{OG} N_{OG} \quad H_{OG} = \frac{V}{K'_y a S} = \frac{V}{K_y a (1-y)_{*M} S} \quad z = H_{OG} N_{OG} = H_{OG} \int_{y_2}^{y_1} \frac{dy}{y-y^*}$$

or

$$z = H_{OL} \int_{x_2}^{x_1} \frac{(1-x)_{*M} dx}{(1-x)(x^*-x)} = H_{OL} N_{OL} \quad H_{OL} = \frac{L}{K'_x a S} = \frac{L}{K_x a (1-x)_{*M} S} \quad z = H_{OL} N_{OL} = H_{OL} \int_{x_2}^{x_1} \frac{dx}{x^*-x}$$

If A is dilute, analytical methods can be applied.

- When $y_{eq}=mx$ and $A=L/mV$:

$$N_{OG} = \frac{1}{(1 - 1/A)} \ln \left[\left(1 - 1/A\right) \left(\frac{y_1 - mx_2}{y_2 - mx_2} \right) + 1/A \right] \quad (22.5-49)$$

$$N_{OL} = \frac{1}{(1 - A)} \ln \left[\left(1 - A\right) \left(\frac{x_2 - y_1/m}{x_1 - y_1/m} \right) + A \right] \quad (22.5-50)$$

The values of m and A to use in the equations above when operating and/or equilibrium lines are slightly curved are discussed in detail in [Section 22.5](#) for use with Eqs. (22.5-4) and (22.5-5).

- If also the operating and equilibrium lines are straight and not parallel:

$$N_{OG} = \frac{N \ln A}{(1 - 1/A)} \quad (22.5-51)$$

- In some cases (GC!) you need to be able to report “height equivalent to a theoretical plate”:

$$\text{HETP} = H_{OG} \frac{\ln(1/A)}{(1 - A)/A} \quad (22.5-52)$$

Get mass transfer coefficients from correlations or measurements.

- Infer height of a transfer unit H_{OG} or H_{OL} from separation measured for a height of packing.
- Single-phase k usually isn't available or meaningful; even overall K has to be measured or inferred from H .
- Very useful correlations: Gas-phase H_{OG} from:

$$H_G = H_y = \left(\frac{0.226}{f_p} \right) \left(\frac{N_{Sc}}{0.660} \right)^{0.5} \left(\frac{G_x}{6.782} \right)^{-0.5} \left(\frac{G_y}{0.678} \right)^{0.35}$$

- And likewise for liquid H_{OL} :

$$H_L = H_x = \left(\frac{0.357}{f_p} \right) \left(\frac{N_{Sc}}{372} \right)^{0.5} \left(\frac{G_x/\mu}{6.782/0.8937 \times 10^{-3}} \right)^{0.3}$$

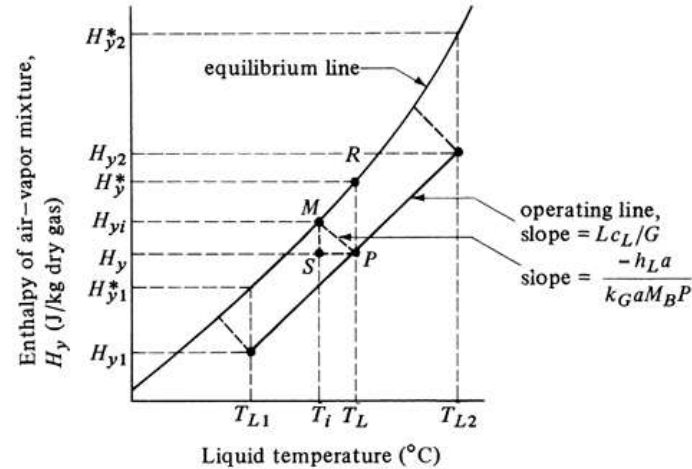
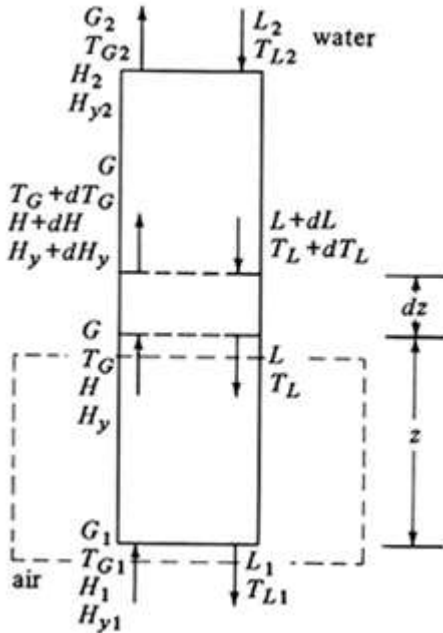
- Giving:

$$H_{OL} = H_L + (L/mV)H_G$$

The big view:

1. Design or analyze these processes: Size the height z (or number of stages) and the diameter (pressure drop) and choose packings.
2. Obtain the needed mass-transfer coefficients.
3. Understand how to use:
 - a) Single-phase mass-transfer coefficients (k_c, k_x, k'_y , etc.)
 - b) Overall mass-transfer coefficients (K_y, K'_y, K_x, K'_x)
 - c) Kremser equations.
 - d) Height of transfer unit and number of transfer units (HTU/NTU) approach.
 - e) Understand to use “A through nondiffusing B” (k_x, K_y , etc.) unless A is very dilute, when equations reduce to “Equimolar counterdiffusion” forms (k'_x, K'_y , etc.)

Remarks on **Humidification**: Continuous countercontacting flow but with heat transfer and with MT only through gas.



$$z = \frac{G}{M_B k_G a P} \int_{H_{y1}}^{H_{y2}} \frac{dH_y}{H_{yi} - H_y}$$

where H_y is enthalpy of air-vapor mixture